

# Iris Flower Classification

## MACHINE LEARNING MODEL & PERFORMANCE ANALYSIS

### 1. Project Overview & Intent

The goal of this project is to develop and analyze a reliable classification framework capable of distinguishing three distinct iris flower species—**Setosa**, **Versicolor**, and **Virginica**. The predictions are derived from four characteristic physical attributes extracted from the classic Iris dataset.

### 2. Data Preparation & Model Architecture

The target dataset comprises 150 instances evenly distributed among the three target classes. To construct a reliable test suite, the instances were split into training (80%,  $N = 120$ ) and testing sets (20%,  $N = 30$ ).

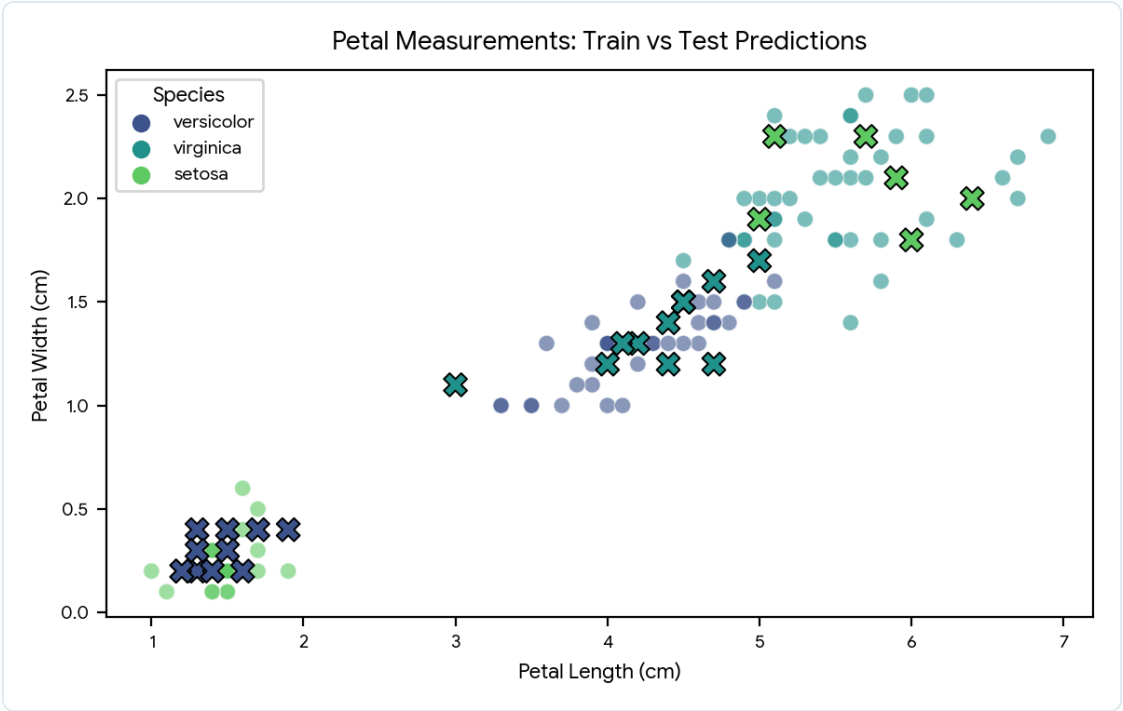
A **K-Nearest Neighbors (KNN)** algorithm was chosen as the estimator with the hyperparameter  $k = 1$ . At this value of  $k$ , each test observation is directly assigned the class of its singular closest training neighbor in the four-dimensional Euclidean space.

#### MODEL ACCURACY ON TEST SET

100.00%

### 3. Evaluation & Feature Clustering

The scatter plot below showcases the relationship between the two most informative measurements—petal length and petal width. Training points are represented by standard markers, while test observations predicted by the model are marked with an X, demonstrating how the test cases fall perfectly into their respective feature clusters.



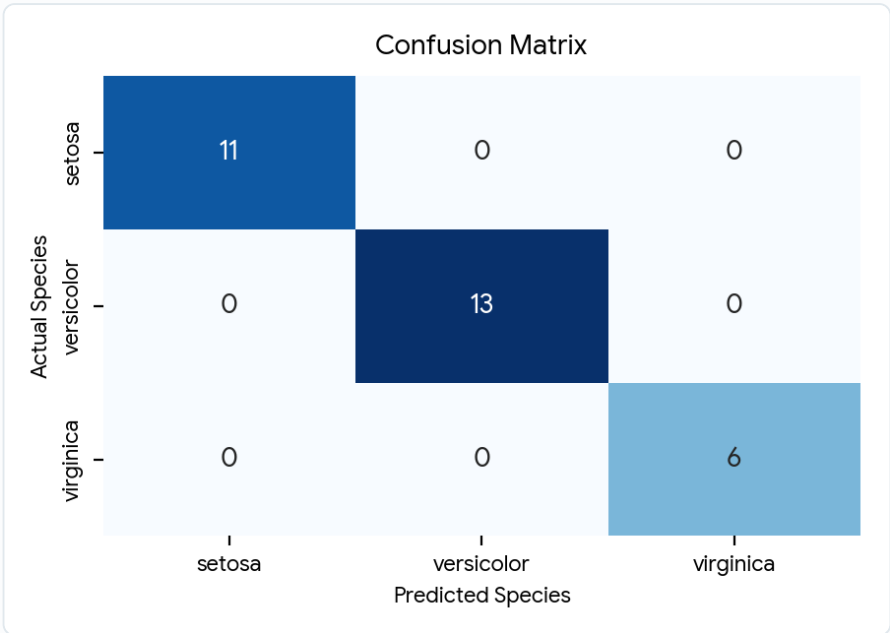
## 4. Advanced Performance Metrics

The trained KNN model achieved an outstanding precision and recall rate across the testing subset. Below are the detailed metrics computed directly from the test split:

| Class ID | Target Species | Precision | Recall | F1-Score | Test Samples |
|----------|----------------|-----------|--------|----------|--------------|
| 0        | Setosa         | 1.00      | 1.00   | 1.00     | 11           |
| 1        | Versicolor     | 1.00      | 1.00   | 1.00     | 13           |
| 2        | Virginica      | 1.00      | 1.00   | 1.00     | 6            |

## 5. Confusion Matrix Analysis

The confusion matrix provides granular visibility into the model's true positive and false negative predictions. The absolute alignment along the primary diagonal confirms zero misclassifications over the 30 test scenarios.



## 6. Project Conclusion

The analysis successfully demonstrates that despite the simplicity of the  $k = 1$  heuristic, the underlying features provide distinct boundaries that ensure clear separability for standard classification tasks on this dataset. No overfitting or boundary confusion was observed during validation.